

VAWSS: Variational Autoencoder-Enhanced Wireless Sensing Simulator for WiFi Channel State Information

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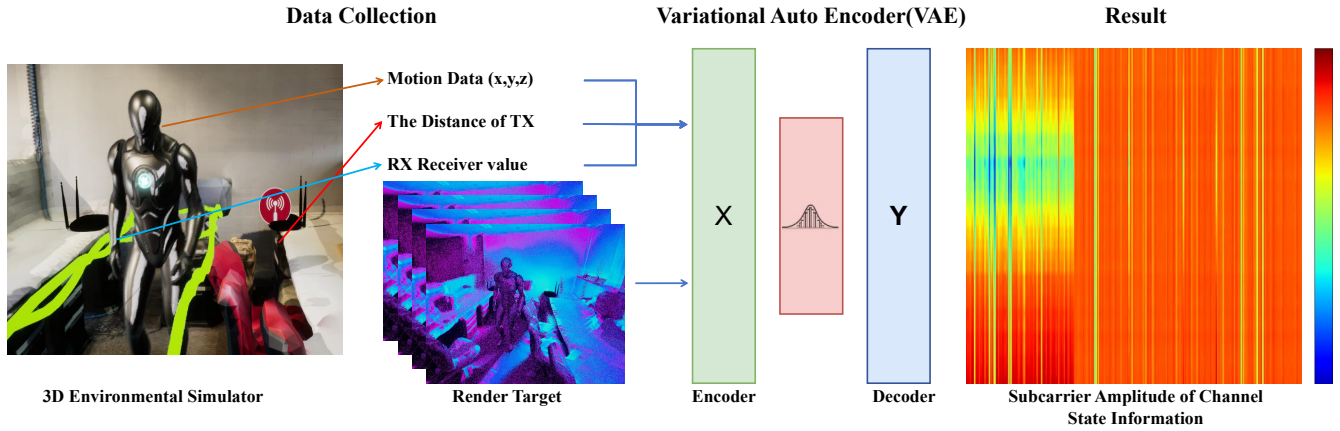


Figure 1: The process of CSI simulation by ray tracing propagation

Abstract

Wireless sensing technology represents a significant milestone in the evolution of modern wireless communication toward sensing capabilities. Current commercial WiFi devices are known to perform various non-contact detections, such as respiration monitoring and gesture recognition, by capturing Channel State Information (CSI). However, the inherent unpredictability of signal fluctuations in real-world environments is a challenge for its application. For example, CSI is highly susceptible to environmental interference in real-world settings, which complicates its usage for the design and analysis of sensing systems. This further leads to limited availability of compatible commodity devices, making the collection of CSI datasets difficult and expensive. Therefore, the deployment of CSI devices and the method of reducing environmental interference present significant challenges in the field of wireless sensing. To

effectively mitigate these issues, we employ GPU-based ray-tracing technology combined with a Variational Auto Encoder (VAE) to simulate CSI variations for analysis. Field experiments have been conducted to evaluate the performance of our simulator.

CCS Concepts

• Computing methodologies → Simulation environments.

Keywords

WiFi Sensing; Channel State Information (CSI); Variational Auto Encoder (VAE)

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1 Introduction

Wireless technologies [7] enhance our lives by enabling communication and connectivity without physical cables. Wi-Fi is one of

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the most widely used wireless communication technologies. Today, with the improvement of various hardware technologies and frameworks, new network devices like high-performance software routers, access points, and access point controllers have redefined the function of wireless communication. These have evolved from merely providing Internet access to becoming a versatile mini-controller equipped with numerous plugins and customizable routing data [11]. Meanwhile, Wi-Fi sensing technology has been applied in numerous scenes. Especially, Wi-Fi Channel State Information (CSI) technology leverages amplitude and phase information to detect object movements and recognize actions [13, 16–18]. However, it is necessary to purchase a specific Wi-Fi network interface card (NIC), such as the Intel WiFi Link 5300, Atheros NIC, or to use professional hardware equipment with openwifi framework such as FPGA [2, 6]. Furthermore, the collected CSI data is highly susceptible to external interference, leading to data distortion and requiring a significant amount of time for analysis.

To address these limitations in real-world CSI collection, we propose the VAWSS system, a wireless sensing simulator based on our digital twin platform WirelessDT [8], which was built on Unreal Engine (UE5) [5]. VAWSS introduces a CSI-based Wi-Fi simulation system, which can simulate 56 subcarriers amplitude for Wi-Fi CSI. It can generate relatively accurate synthetic CSI subcarrier amplitude data in real-time using the GPU-based ray tracing technology and a Variational Autoencoder (VAE) [12].

Specifically, we developed a lightweight multidimensional input VAE model to encode and decode variables by using ray tracing, render targets (texture widget in UE5), object motion, and the distance of the target from the transmitter. VAE has been widely used in sensor networks and data generation tasks because of its capability to model complex data distributions. Previous research has demonstrated the effectiveness of VAE in various applications, such as anomaly detection in sensor networks, image generation, and data compression. In the context of sensor networks, VAEs are utilized to learn latent representations of high-dimensional sensor data, enabling efficient data generation and reconstruction. We leverage the VAE model for generating the CSI subcarrier amplitudes. By extracting latent features from the CSI data, we aim to understand better and analyze the underlying channel characteristics. The VAE generative process enables the creation of realistic CSI subcarrier amplitude data, which can be used to simulate various communication scenarios.

The contributions of this work include the proposal and implementation of the first real-time CSI subcarriers simulation system for Wi-Fi signals. Our system contributes to a comprehensive understanding of wireless channel behavior and enables the development of inventive solutions to modern challenges in wireless networks. In addition, Wi-Fi CSI subcarriers contain both amplitude and phase information [1]. The VAWSS system focuses primarily on amplitude for model training and data generation. The process of VAWSS system is illustrated in Figure 1.

2 Related Work

Currently, there are numerous simulation and modeling methods used to evaluate radio electromagnetic wave scenarios in environments. Cisco DNA Center [15] is a platform for calculating Received

Signal Strength Indication (RSSI) data in 2D and 3D buildings, conveniently mapping heatmaps onto 2D and 3D models. Cisco focuses primarily on signal strength systems that confirm signal strength between buildings. In contrast, Carla and AirSim [4, 14] focus on self-driving systems, using ray tracing to simulate visual signals, for example, realistic senses, but their support to simulate various wireless sensors and signals, for example, WiFi, is limited. Furthermore, these simulators cover communication and sensing technologies in the wireless field, but they are not suitable for real-time high-resolution computation. In addition, these platforms are designed for specialized areas such as buildings and automobiles. Thus, they are not directly applicable to wireless sensing assessments in high-resolution dynamic scenarios. Other systems include AURELION Sensor Simulation and Wireless InSite 3D Wireless Propagation Software [9]. Both platforms support large-scale environment modeling and ray tracing based propagation simulations [3]. Compared to these systems, the Variational Autoencoder-Enhanced Wireless Sensing Simulator (VAWSS) offers multi-interfaces for the user. VAWSS provides more detail in wireless propagation simulations, ranging from MAC layer-based RSSI to PHY layer-based CSI information, enabling comprehensive CSI subcarrier simulation and testing.

3 Implementation

To achieve a better simulation effect, we conducted a 3D reconstruction of an office and imported the environment model into UE5, where we performed wireless modeling of the environment, GPU ray tracing simulation, mobile object tracking, transmitter (TX) distance acquisition, receiver (RX) prediction, and VAE model simulation and generation. The overall architecture of the system is illustrated in Figure 2.

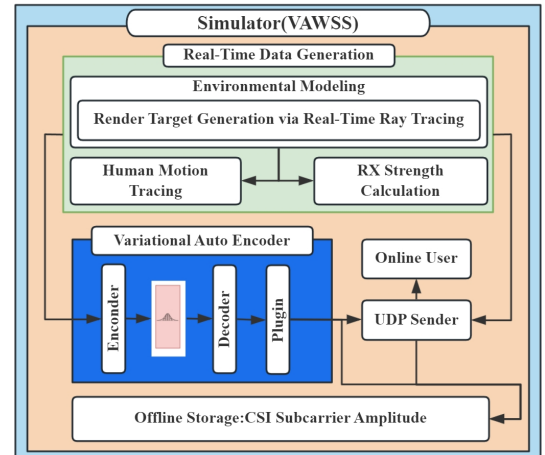


Figure 2: Architecture Framework

3.1 Real-Time Data Generation

To efficiently deploy the environmental model and generate an environmental render target, we use the UE5 rendering tools and the blueprint tools to develop the render target with the motion

of the object. Using NVIDIA's Real-Time Ray Tracing (RTRT) [8], UE5 reconstructs the wireless signal propagation process and sends the results to the VAE unit for encoding. In this section, we use a human model as the moving object for our experiments.

- **Environmental Modeling:** The GPU data generation processes include three steps: 1) create a scene map using a Texture Render Target to generate a small-scale map; 2) utilize the GPU to perform calculations based on the mapped pixels, and generate the mapped ray-tracing diagram in an octahedral mapping [10] to the transmitter and receiver antennas; 3) output the ray information captured in the scene, and extract the signal strength at the hit points. This comprehensive data is ultimately stored in a 2D image cache for further analysis. This method ultimately contributes to a more robust and reliable sensing system, as illustrated in Figure 3.
- **Render Target Generation via Real-Time Ray Tracing:** Render target is a texture widge in UE5. It consists of a small map of the real-time simulation engine. The target render is created by ray tracing points in our system, which is shown in second step of Figure 3. To enhance the training process of the model, we use different depths of color to represent signal values, which created by the count of ray tracing. This approach allows for a more detailed representation of signal strength variations, facilitating more accurate and efficient training. By incorporating a range of color depths, the model can better distinguish between different signal intensities, leading to improved performance and more precise predictions. The results of this method are presented in the third step of Figure 3.
- **Human Motion Tracing and RX Strength Calculation:** We set the moving object in the VAWSS system as a human character. This object walks along the same path as in reality while outputting its position information. These include the object's coordinates in the world coordinate system, the absolute distance between the object and the TX transmitter, and counting the intensity and number of ray tracing points falling on the human body that calculate the RX strength value of the object's current location.

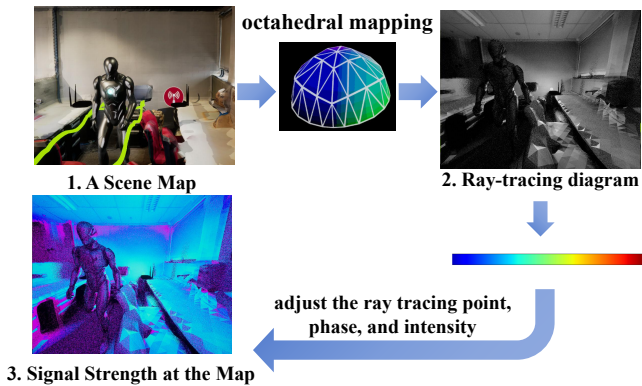


Figure 3: Process of render target generation

3.2 VAE Data Simulation

To achieve simulation data that closely approximate real-world distributions, we employ a Variational Autoencoder (VAE) model for data fitting and generation. The VAE model can extract latent features from CSI data, facilitating understanding and analysis of channel characteristics. Additionally, the model uses Nvidia GPU Compute Unified Device Architecture (CUDA) technology to accelerate processing, improving real-time performance.

- **VAE Model:** The architecture of VAE, encoder-decoder architecture, compresses high-dimensional ureal engine data into a low-dimensional latent space and then generates new data from this space. In this process, we require the input data to include the target values of the image, and the movement of the human (x , y , z), where x and y represent the position of the human within the model, and the z -axis indicates the height, along with the relative distance from the transmitter (TX) to the human. The encoder and decoder architecture not only retain the primary features of the data, but also reconstruct 56-dimensional CSI subcarriers similar to the original data through the decoder. This method effectively captures and generates complex channel states, enhancing understanding of channel characteristics.
- **Plugin for GPU Inference:** We design the VAE model to perform inference using Nvidia CUDA cores. First, we implement the VAE model with CUDA support to ensure efficient processing. Next, we utilize the Unreal Engine 5 (UE5) plugin mechanism to cast and transmit the data from the simulation environment to the VAE model. This integration enables data stream and high-performance inference, allowing the VAE to process CSI data in real-time, capturing and generating complex channel states for enhanced understanding of channel characteristics.
- **UDP Interface of CSI Subcarriers:** To rapidly export CSI subcarrier data, we use the UDP protocol to transmit the amplitude data with the timestamp of the CSI subcarriers. Users can set up a UDP client to connect to the UDP server, facilitating real-time data transfer. The generated data will include the coordinates of moving objects, allowing users to track and analyze movement within the simulated environment. This setup ensures efficient and seamless data communication, allowing detailed analysis and visualization of CSI subcarrier variations in response to changes in the environment. Additionally, we introduced UDP transmission for these training data, allowing developers to easily export and test the data in the early stages.

4 Evaluation

4.1 Experimental Environment Setup

In the physical world, we used the Atheros CSI tool for data collection, utilizing an Intel Celeron N4505 and an AR9382 NIC for hardware collection. In simulation, the machine used to run the VAWSS system is illustrated in the Table 1. This setup includes detailed specifications of the hardware and software components essential for the effective operation of the VAWSS system. These specifications ensure that the system is capable of handling the computational demands of real-time CSI subcarrier amplitude data generation and processing.

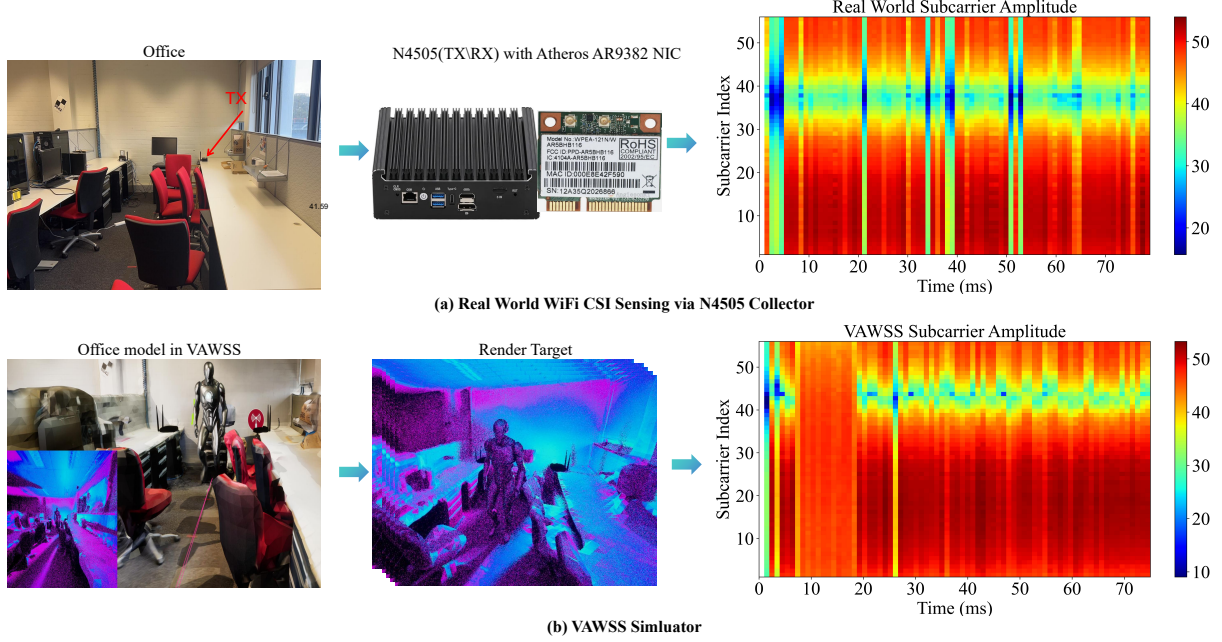


Figure 4: The comparison between real-world data and simulation system data.

4.2 Physical Data Collection

We proposed an experiment to collect the accuracy CSI data in real world for VAWSS training. Four participants walked along a fixed path in the office, collecting data from different positions. Physical markers were placed on the floor every 1 meter to indicate that a fixed path segment of 1 meter had been walked. Each of the four participants walked back and forth along the path, completing six rounds in total. The real world data show in Figure 4 (a). In Figure 4, the Y-axis represents the subcarrier index (the total of 56 subcarriers), the X-axis indicates time, and the color represents the amplitude strength.

Table 1: VAWSS Deployment Environment.

	Specification
Simulator Hardware	CPU: Intel(R) i5-12400, GPU: RTX 3070 Ti, Memory: 16 GB DDR4 RAM
Collector Hardware	CPU: Intel Celeron N4505, NIC: AR9382
Simulator Software	OS: Windows 11 Pro, CUDA Version: 12.5, Unreal Engine: 5.0.2
Collector Software	OS: Ubuntu 14, Kernel: 4.1.10, Memory: 8 GB DDR4 RAM

4.3 VAWSS System Performance

The amplitude data from the real world and the VAWSS system are shown in Figure 4. In the VAWSS system, we conducted a high-precision simulation of a real office environment to ensure that the ray tracing system closely approximates the physical environment during operation. To evaluate the VAWSS platform, the moving

object walks along a path similar to the real-world trajectory and collects data accordingly. We sampled the subcarrier amplitude of the corresponding index every 10 ms, compared the amplitude of the real collected data with the VAWSS data, calculated the amplitude strength of the subcarrier of the corresponding index in this time period, calculated the error in the total time, and averaged the total error. The latency for generating data within the entire VAWSS system framework is approximately 83ms. The performance of the VAWSS system is shown on Table 2.

Table 2: Performance Metrics of the VAWSS

VAWSS System	
Error rate in VAWSS	26%
FPS of CSI data generation	12±3
VAWSS latency	83ms

5 Conclusion and Future Work

In this study, we designed and implemented VAWSS, a wireless sensing simulator for WiFi CSI, and used a Atheros NIC to collect and analyze the CSI measurements in real-world experiments to evaluate its performance. Currently, our design primarily focuses on generating amplitude data, as amplitude variations and distributions are easier to observe. However, we are unable to address the issue of non-linear phase error in CSI. In addition, to ensure real-time data processing and accuracy, we designed a relatively lightweight VAE model. This approach is necessary because the render target sampling itself is computationally intensive, limiting the processing power to 12 FPS. In the future, we will enhance the functionalities of VAWSS and optimise its system performance.

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